

QuickGrade: An Interactive Multimodal Essay Feedback System for K-12 Writing Education

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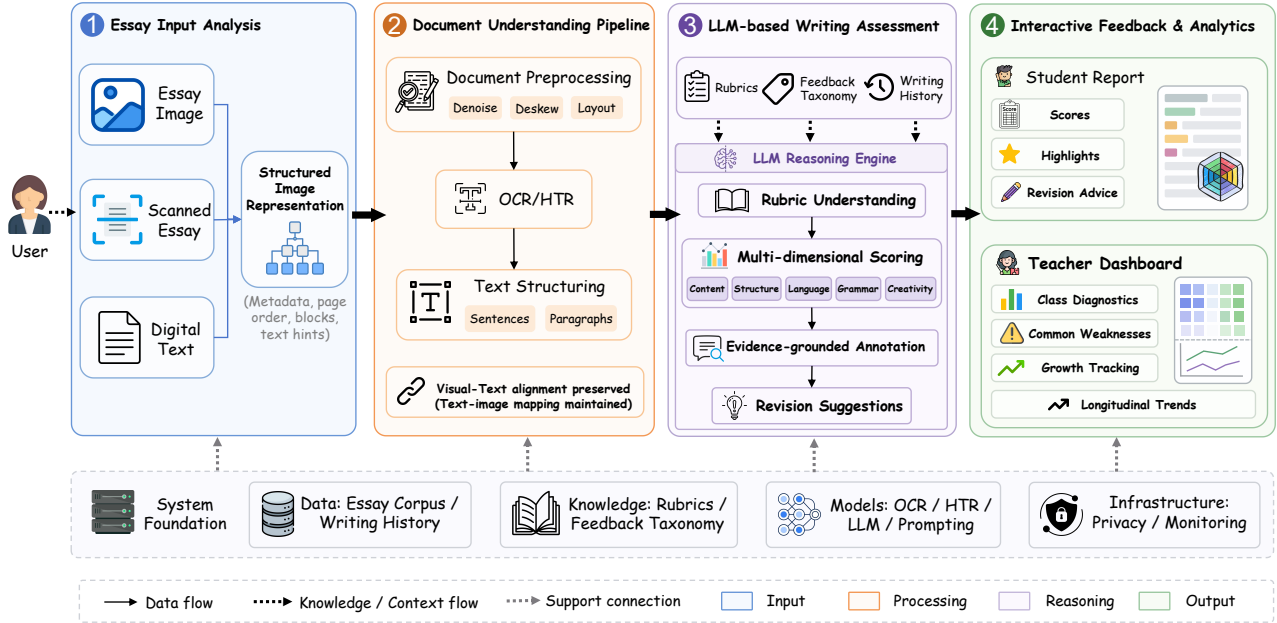


Figure 1: System architecture of QuickGrade.

Abstract

Writing assessment in K-12 education is labor-intensive, requiring teachers to evaluate content, organization, logic, language, creativity, and grammar while providing actionable feedback. Existing automated essay scoring tools mainly handle clean digital text, limiting support for handwritten, photographed, or paper-based submissions. We present QuickGrade, an interactive multimodal essay feedback system for classroom writing instruction. Given a photographed essay, scanned document, or digital text, QuickGrade normalizes the document, reconstructs sentence- and paragraph-level structures, and generates rubric-based scores, evidence-grounded comments, and revision suggestions. Through an interactive interface, students can inspect highlighted sentence-level feedback, while teachers can review individual reports, class-level patterns, and longitudinal progress. By supporting a complete workflow from

essay submission to feedback generation and teacher-facing analytics, QuickGrade demonstrates how multimodal AI can enable scalable and teacher-in-the-loop writing instruction. Supplementary materials are available at: [Google Drive folder](#)

CCS Concepts

• **Applied computing** → Education; • **Human-centered computing** → Interactive systems and tools; • **Computing methodologies** → Natural language processing; Computer vision.

Keywords

Automated Essay Assessment; Multimodal Learning Analytics; Writing Feedback; Document Image Understanding; Large Language Models; Interactive Educational Systems

1 Introduction

Writing assessment is a central yet labor-intensive component of K–12 education. Teachers must evaluate not only grammar and spelling, but also content relevance, organization, logic, expression, and creativity. When large numbers of handwritten or photographed essays need to be graded, feedback is often delayed, inconsistent, or difficult to personalize.

Automated essay scoring (AES) has been studied for decades, from early statistical methods to e-rater and neural scoring models [1, 3, 6, 8]. Recent large language models further enable flexible writing feedback and reasoning over text quality [2, 10]. However, most existing tools assume clean digital text input and provide limited support for paper-based classroom scenarios. Moreover, a holistic score alone is insufficient for writing instruction; students and teachers need localized, explainable, and actionable feedback grounded in the essay.

To address these limitations, we present QuickGrade, an interactive multimodal essay feedback system for K–12 writing education. QuickGrade supports paper-based and digital essay submissions and provides an end-to-end workflow for transforming classroom writing materials into explainable feedback and learning analytics. The system combines document understanding, rubric-guided LLM assessment, and interactive feedback visualization, enabling students to inspect localized comments and teachers to diagnose individual and class-level writing patterns. QuickGrade illustrates how multimodal AI can support scalable, transparent, and teacher-in-the-loop writing instruction in realistic classroom scenarios.

2 System Workflow

QuickGrade is organized as an end-to-end workflow that connects essay submission, automated assessment, and interactive feedback. In a typical classroom scenario, a teacher or student uploads a handwritten, photographed, or digital essay. The system first normalizes the document and reconstructs its content into sentence- and paragraph-level structures, preserving the alignment between recognized text and the original essay image. Users can inspect and edit the recognized text before assessment, ensuring accuracy and transparency. A rubric-guided LLM assessment engine then generates dimension-level scores, evidence-grounded annotations, and revision suggestions. Students can explore highlighted feedback on individual sentences, while teachers can switch to dashboards that summarize student performance, class-level patterns, and longitudinal writing trends. The workflow is organized around three main components that match the system architecture in Figure 1. The Document Understanding Pipeline handles essay input, preprocessing, and text reconstruction. The LLM-based Writing Assessment engine generates rubric-guided scores and evidence-grounded annotations, while the Interactive Feedback and Analytics module presents feedback to students and dashboards to teachers for revision, diagnosis, and class-level performance tracking.

2.1 Document Understanding Pipeline

The document understanding pipeline converts submitted essays into analyzable text. It accepts multiple input formats, including handwritten or photographed paper essays and digital text files. The

pipeline performs denoising, deskewing, layout analysis, text detection, OCR or handwritten text recognition (HTR), and sentence-paragraph reconstruction. Inspired by recent document AI and OCR models [4, 5, 11, 12], QuickGrade preserves the mapping between recognized text and its original visual location. This visual-textual alignment allows students and teachers to compare the original essay image with the recognized content and enables subsequent modules to generate evidence-grounded feedback directly linked to the reconstructed text.

2.2 LLM-based Writing Assessment

Using the reconstructed essay produced by the Document Understanding Pipeline, the LLM-based assessment engine evaluates the writing according to grade-level rubrics. The engine produces dimension-level feedback for content, structure, language, grammar, and creativity. Each score is paired with evidence from specific sentences or paragraphs, allowing students to see which parts of their essay contribute to strengths or weaknesses and enabling targeted revision. This sentence-level and paragraph-level grounding ensures that the feedback is interpretable and directly linked to the essay content. This grounding allows the Interactive Feedback and Analytics module to present evidence-grounded annotations for students and diagnostic dashboards for teachers.

2.3 Interactive Feedback and Analytics

The outputs from the LLM-based Writing Assessment module are presented through an interactive feedback and analytics interface. For students, QuickGrade organizes the assessment results into word-level corrections, sentence-level comments, paragraph-level suggestions, and essay-level summaries. Users can click highlighted text to inspect the corresponding explanation, evidence, and revision advice, making the feedback directly traceable to the original essay. For teachers, the system extends individual feedback into student reports and class-level dashboards. These views summarize recurring weaknesses, score distributions, and longitudinal writing progress, supporting data-driven writing instruction [7, 9]. Rather than replacing teachers, QuickGrade is designed as a teacher-in-the-loop assistant that reduces repetitive correction work and helps teachers focus on higher-level guidance, classroom diagnosis, and targeted writing instruction.

3 Expected Impact and Conclusion

QuickGrade presents an interactive multimodal essay feedback system that brings automated writing support closer to realistic K–12 classroom practice. Instead of assuming clean digital text, the system supports paper-based, photographed, scanned, and digital essays, and turns them into explainable feedback through document understanding, rubric-guided LLM assessment, interactive annotation, and learning analytics. By linking sentence-level evidence with student-facing feedback and teacher-facing dashboards, QuickGrade provides timely revision guidance for students and actionable diagnostic evidence for teachers. The system illustrates how visual document processing and language reasoning can support teacher-in-the-loop writing instruction as a practical example of human-centered multimedia technology for education.

Acknowledgments

The authors thank their colleagues for helpful discussions and feedback during the development of QuickGrade, and the anonymous reviewers for their constructive comments that improved this work.

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